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Current trends in the treatment of ocular melanoma by radiotherapy

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ABSTRACT

The Collaborative Ocular Melanoma Study (COMS) has recently confirmed once and for all that it is safe to attempt to preserve an eye with a posterior uveal melanoma by demonstrating no survival advantage of enucleation over plaque radiotherapy. While COMS has been under way, we have set out in London to define the selection criteria for conservative therapy versus enucleation for the various categories of melanoma in terms of size, location within the eye, and presence or absence of retinal detachment. The evolution of this approach has culminated in an overall ocular survival rate of 94% in 597 patients following radiation therapy combined with a mean loss of visual acuity of only 2.4 Snellen lines in the eyes preserved.

Key words: plaque therapy, posterior uveal melanoma, proton therapy, radiotherapy.

INTRODUCTION

History of radiotherapy for ocular melanoma

Radiation therapy for uveal melanoma was introduced in London in the late 1920s. At that time radiotherapy was relatively primitive and precise teletherapy techniques had not yet been devised. Orthovoltage X-ray therapy was available but it was not until 1937 that megavoltage techniques were to be introduced to the world by the Radiotherapy Department of Saint Bartholomew's Hospital. Even these were unsuitable for precision radiotherapy to small targets and brachytherapy methods were still in widespread use. The first successful radiotherapy treatment of a melanoma was carried out on 18 February 1929 at Saint Bartholomew's Hospital by Robert Foster Moore, who inserted a radon seed into a melanoma within the eye,¹ a technique that with Hyla Stallard, his then registrar, he was also exploring in retinoblastoma. The radon was left in place to extinction but the impracticalities of intraocular insertion were soon replaced

by embedding the seeds in wax, positioning them on the surface of the eye, and later removing them. A search by Stallard and his physicist, George Innes, for a simple and reproducible surgical brachytherapy technique culminated in the introduction of the cobalt-60 plaque. By 1966, Stallard was ready to report the results of treating his first 100 cases of melanoma and to define the tumour selection criteria for the method.² Stallard's successor, Michael Bedford expanded our knowledge of the use of cobalt plaques to treat uveal melanoma.³

It was not until the 1970s that the first simulation and animal experiments in charged particle therapy for the eye were conducted by Ian Constable working as a fellow in the retinal service at the Massachusetts Eye and Ear Infirmary in Boston and using the Harvard cyclotron.^{4,5} It was left to his successor, Evangelos Gragoudas, to develop with Goitein and Koehler a practical technique using a proton beam to treat ocular melanomas.⁶ In San Francisco, Char and Castro began to use helium ions for the same purpose.⁷ Following the publication of encouraging results using protons and helium ions in the USA, we decided to evaluate this method in London and to compare it with plaque brachytherapy. Proton therapy in the UK began in 1989.

Throughout this period, radiotherapy remained a controversial method employed in a very few centres in the world for the treatment of a small minority of patients with uveal melanoma. Nevertheless, the attraction of retaining the eye, often with useful vision, led to a slow but steady increase in the number of patients treated in London from two or three cases a year in Stallard's time to 15 or so during Bedford's tenure, and compared with around 180 now. The high energy of the 1.3 MeV emissions from cobalt-60 meant that plaques made with this isotope could not be shielded on their external surface. The side-effects of radiation scatter from high-energy cobalt plaques on the adjacent retina, choroid and optic nerve for posterior melanomas, and on the lens, lids and lacrimal apparatus for anterior tumours led to the development of low-energy brachytherapy, first with ruthenium-106 and then with iodine-125 applicators,^{8,9} both of which could be shielded.

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Principles of radiotherapy for ocular melanoma

Melanomas are not radiosensitive and a tumour necrosis rather than a therapeutic dose of radiation far in excess of the tolerance of the globe is required to destroy this neoplasm within the eye. Plaques rely on the effect of attenuation of radiation in accord with the inverse square law to spare adjacent ocular structures in a way that cannot be achieved with conventional external beam radiotherapy. This effect is progressively lost with increasing tumour thickness and it has been suggested that an ideal dose distribution cannot be achieved for tumours close to the optic disc. Charged particles, both protons and helium ions, can be directed in a pencil beam in which the dose falls rapidly at the edge so that adjacent tissues can be preserved. Furthermore, the Bragg peak effect of charged particles means that the beam cuts off completely behind the treated field so that structures behind the eye are totally spared. This Bragg peak also offers a significant reduction of the entry dose so that, to a degree, the eyelids can be spared, although this effect is progressively lost with more anterior tumours and with modulation of the beam to treat thicker tumours. Unlike plaques, in which the dose falls steadily from the scleral surface to the tumour apex, protons and helium ions offer the opportunity to achieve a physically (though not radiobiologically) uniform dose of radiation within a tumour. Thus charged particles seemed to offer an ideal dose distribution to treat a radioresistant tumour like melanoma within a radiosensitive structure such as the eye, particularly when the neoplasm was too large or too close to the optic disc to treat by plaque. With time and the opportunities offered by new methodology it became increasingly apparent that two key questions had to be answered.

Is radiotherapy a safe alternative to enucleation?

At the end of the 1970s, and soon after the introduction of these newer approaches, conservative treatment for ocular melanoma was brought into prominence by the opposing views of two ocular pathologists. Zimmerman published his opinion that, far from reducing the risk of dissemination, removing an eye with a melanoma might actually potentiate metastatic disease.¹⁰ In contrast, Manschott ventured to suggest that treatments that aimed to conserve the eye might themselves precipitate distant spread.¹¹ This Catch 22 controversy posed serious difficulties for the clinical practitioners of ocular oncology at that time. It ran counter to their experience from large specialist centres where no difference could be found in survival rates when the outcomes of radiotherapy and enucleation were compared. In Philadelphia, Augsburger had found no significant difference in survival between patients treated by plaque and by enucleation.¹² Similarly, no difference was reported from a study of plaque data combined from New York City and Saint Bartholomew's Hospital, London.¹³ From Boston came a report that there was little difference in the long term between survival rates after proton beam radiotherapy and

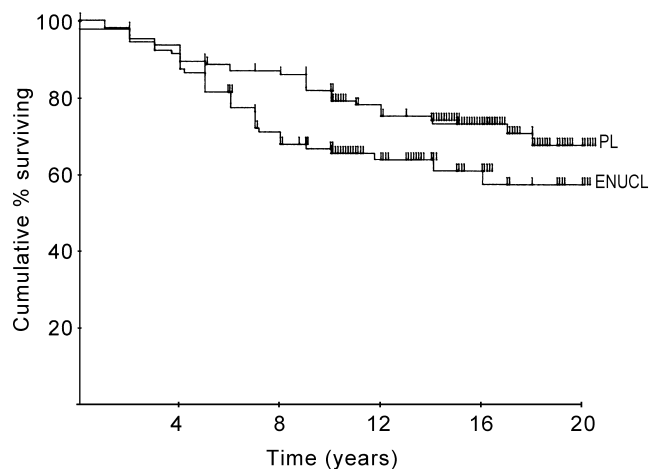


Figure 1. Kaplan–Meier survival curves comparing primary plaque radiotherapy ($n = 123$) and enucleation ($n = 104$). $\chi^2 = 4.035$; $P = 0.045$.

enucleation.¹⁴ All these studies suffered from the disadvantages of relatively short follow-up periods with medians, where reported, of around 5 years, and of not being randomized.

In 1985 this author's unpublished retrospective, data from Moorfields Eye Hospital, London, compared survival rates between 123 consecutive Oncology Service patients treated by primary cobalt plaque therapy alone with 104 individuals from other services within the Hospital undergoing primary enucleation and matched for age, sex, and the size and location of their melanoma. The standard dose, introduced by Stallard and Innes, of 80 Gray (Gy) at the design depth of the applicator was used for all patients in this series. The median follow-up period for both groups was 16 years and, again, comparing Kaplan–Meier curves, there was no significant difference between plaque and enucleation (Fig. 1). There was no bias of selection between the two groups; however, although follow-up was much longer, this study still suffered from possible errors in achieving a precise match, particularly of tumour size in the pre-ultrasound era. It was for this reason that the series was not prepared for publication.

Later in 1985, the potential for inaccuracies of this type was used to justify funding for a multicentre, prospective, randomized trial, the Collaborative Ocular Melanoma Study (COMS). By this time the use of cobalt plaques was in decline and it was decided that the main study should compare survival rates after iodine-125 plaque therapy with those after enucleation. The results of this comparison of conservative with radical treatment for posterior uveal tract melanoma were not published until the summer of 2001,¹⁵ albeit again with a short follow-up of only 5 years. The final data from this lengthy and expensive study that involved the enucleation of 660 eyes to be balanced against conservative treatment of 657 bear remarkably close comparison with the non-randomised, retrospective Moorfields data from 1985, particularly with regard to the conservative option (Table 1). It was plaque therapy that was under critical evaluation and

Table 1. Comparison of Collaborative Ocular Melanoma Study (COMS) and Moorfields 5-year survival rates

Therapy	Moorfields (1985) No. patients (%)	COMS (2001) No. patients (%)
Plaque	112/123 (91)	598/657 (91)
Enucleation	89/104 (86)	587/660 (89)

Table 2. Visual results of cobalt-60 plaque therapy (median follow up 16 years) in eyes retained in surviving patients at end of study period

Visual acuity	No. patients (%)
>6/12	25/98 (25)
6/60–6/18	34/98 (35)
<6/60	39/98 (40)

5-year survival rates after brachytherapy were identical at 91% in both the Moorfields and COMS investigations. The 5-year survival rate after enucleation in the COMS study was 89%, also comparing well with the Moorfields value of 86% and with the other earlier series. This illustrates that data from carefully matched retrospective analysis should not necessarily be dismissed in favour of prospective information, particularly when the proof may require that a large number of individuals is subjected to a potentially unnecessary radical procedure. The prospective COMS study thus confirmed the finding of older, retrospective studies that radiotherapy was indeed a safe alternative to enucleation.

What is the best radiotherapeutic approach?

Opinion in the large European centres, based on retrospective data, was that COMS was unlikely to show a substantial difference between plaque and enucleation and this seems to have been borne out in practice. Consequently, none of these centres took part. More particularly, in London, it was felt that any difference that was shown would be insufficient to change clinical practice, especially when this would mean moving from an attractive method that conserved the eye to a radical one that sacrificed it. The 15 years spent on randomizing patients in the USA were therefore devoted to an attempt to devise a structured therapeutic approach using more than one radiation modality. The intention was to define the interfaces between simpler, milder, more cost-effective techniques with less side-effects using radioactive plaques and the more aggressive, expensive methods with more adverse effects employing charged particles.

PLAQUE RADIOTHERAPY

Our review of cobalt plaque data had shown that 80% of the surviving patients carefully selected for this therapy according to the criteria of Stallard and Bedford retained the

Table 3. Comparison of ocular retention rates after low-energy treatments (median follow up 3 years)

Treatment	No. eyes retained (%)
Ruthenium-106	93/106 (88)
Iodine-125	85/90 (94)

Table 4. Comparison of visual outcomes at 6 years after low-energy treatment, where visual loss was defined as loss of visual acuity >2 Snellen lines

Treatment	No. with visual loss (%)
Ruthenium-106	31/106 (29)
Iodine-125	47/90 (52)

treated eye at the end of the study. Table 2 sets out the visual results in these retained eyes, which depended largely, of course, on the location of the tumour in relation to the optic disc and macula. Despite criticism of the high activity of cobalt plaques, 60% of patients kept a visual acuity of 6/60 or better, often after very long periods of time. By 1985, cobalt plaques were no longer in regular use in London and began to be replaced by shielded ruthenium and iodine applicators. We thought that these low-energy sources might have a reduced radiobiological effect compared with their high-energy predecessors and, empirically, increased the dose at depth to 100 Gy. By now, echography was in regular use to estimate accurately the thickness of the melanoma, and the dose was prescribed at the depth of the tumour apex from the surface of the eye including a nominal 1 mm for the scleral thickness. This method of dosimetry replaced the older 'design depth' method employed in the pre-ultrasound cobalt era.

By 1991 we were in a position to evaluate our first results of the newer low-energy approach. One hundred and six patients with relatively thin melanomas up to 5 mm in elevation were treated with ruthenium plaques emitting mainly non-penetrating beta rays and 90 thicker tumours elevated up to 8 mm with iodine applicators generating more penetrating X-rays. These two groups of patients were therefore selected on the criterion of tumour thickness. Follow up was complete for the 196 subjects. Table 3 compares the ocular retention rates that were found to result from the two low-energy treatments after a median follow up of 3 years (range 1–6 years). It was immediately apparent that, paradoxically, more eyes with the thicker tumours treated by iodine plaques were preserved than with the thinner lesions for which ruthenium was used. Table 4 compares the visual outcomes. Loss of visual acuity of more than two Snellen lines compared with the preoperative value was chosen as the outcome measure as it corrects best for tumour location. It was quite clear that the less penetrating radiation from ruthenium plaques produced far less visual side-effects.

Table 5. Comparison of ocular retention rates after ruthenium beta radiotherapy and dose modification with earlier results of X-ray therapy with iodine (median follow up 3 years for both methods)

Treatment	No. eyes retained (%)
Dose-modified ruthenium-106	91/98 (93)
Iodine-125	85/90 (94)

The next step in the evolution of brachytherapy in London was to determine whether altering the dose of radiation delivered by ruthenium plaques could emulate the better local tumour control rates achieved by iodine sources while, at the same time, maintaining the ability of the beta-emitting applicators to preserve better vision than those employing X-rays. Dose modification was achieved in two ways, first by increasing the depth dose in the tumour and second by increasing the size of the applicator used. The normal depth dose of 100 Gy was empirically increased to 120 Gy for tumours more than 5 mm thick. It was decided not to grade this higher dose according to increasing depth over the 5 mm limit as was done for tumours below this thickness. This decision was taken because the surface dose from ruthenium applicators rises much more steeply with increasing depth dose than does that from iodine plaques and we feared scleral necrosis. Consequently, a group of patients with tumours greater than 5 mm and up to 8 mm in thickness began to be treated with ruthenium plaques to a dose of 120 Gy routinely prescribed at a depth of 5 mm.

We had noticed in clinical practice that most plaque failures were due not to inadequate regression of the tumour as a whole but to edge recurrence. This adverse effect was much more common following ruthenium treatments than after iodine or, indeed, cobalt plaque therapy. We concluded that, although the reduced scatter from beta sources brought with it the advantage of less damage to the disc or macula when they were not directly involved by tumour, it had to be set against the disadvantage that the safety margin at the edge of a melanoma was significantly less than with more penetrating X-ray or gamma emissions, particularly for lesions with relatively indeterminate margins. We countered this effect by increasing the use of ruthenium applicators of 20 mm nominal diameter relative to that of the 15 mm applicators that we had mostly used in earlier studies.

Table 5 compares the initial results of dose modification in 98 patients having a median follow up of 3 years (range 1–4 years) with the earlier results from Table 3, also at a median follow up of 3 years, and shows how this technique has increased the local tumour control rate to be more comparable with that achieved by iodine applicators. The initial impression is that this benefit has occurred without a significant adverse effect on the visual advantages of ruthenium plaques, although this requires formal evaluation. Furthermore, patients treated by ruthenium plaque did not develop cataract whereas those in whom iodine applicators were used often had to undergo cataract extraction. These

preliminary results have led us away from iodine plaques, which had particularly severe side-effects when used to treat thicker melanomas.

Proton beam radiotherapy

By 1992 we were ready to begin evaluating the results of 6 years of proton treatments of London patients. Our early experiences of this method had been singularly disappointing. We had used protons exclusively to treat melanomas that were too large or too close to the optic disc to employ plaque brachytherapy and had observed a high incidence of complications. In common with other types of radiotherapy, many of these were not apparent for 2–3 years after treatment. The dose distribution characteristics of the modulated proton beam led us to suspect that the lids, external eye and anterior segment would receive a high dose of radiation when the method was used to treat thick tumours and those involving the ciliary body, and this was borne out in practice. Adequate retraction of the eyelids out of the beam during treatment proved difficult to achieve, particularly in young people with no lid laxity. As a result, marked loss of lashes and lid margin keratinization were observed. Cosmetically unacceptable pallor of the eyelid skin was seen in patients with a dark complexion. These problems did not occur with plaques. Their consequences were most marked in the upper eyelid and were so prominent after protons that we decided early on not to treat superior, anterior melanomas by this method.

At about the same interval after proton therapy we began to see rubeotic glaucoma. This was most marked in patients with large tumours who had had persistent subretinal fluid after treatment and we realized that it was less a direct late effect of radiation than it was the result of ischaemic retinal detachment. In turn, this depended on another effect we had observed, which was that tumour regression and reabsorption of subretinal fluid was much slower after protons than after plaque for a given size of tumour. In effect, rubeotic glaucoma was common after protons because the physiology of the eye was either slow to or never returned to normal. These bad experiences led us to treat larger tumours and those next to the disc by plaque using an increased dose where appropriate. To be fair to protons, the plaque approach also produces some cases of rubeotic glaucoma and we are in the process of comparing the incidence of rubeosis after plaque and protons for comparable melanomas.

The late effects we have described were not widely reported in other series and there were two possible reasons why this might have been so. First, there were differences in the cyclotrons used to deliver protons. In the UK, we used a small machine designed for medical treatment and working at its full 65 MeV capacity. Elsewhere in the world, powerful old cyclotrons originally designed for physics research were being used. The protons from these machines had to be filtered through range limiters effectively to

Table 6. Relationship of tumour size and retinal detachment to rubeotic glaucoma following proton therapy

Feature	No. eyes developing rubeotic glaucoma (%)
Tumour too large to plaque and presence of retinal detachment	35/40 (88)
Tumour too large to plaque <i>or</i> presence of retinal detachment	19/52 (37)
Neither feature	3/35 (9)

reduce their energy and therefore their penetrance. It is possible that, by altering the radiobiology of the proton beam, this trick of physics may reduce side-effects. The second and more compelling reason was that most of the other centres using charged particle therapy were not also using plaques for the smaller and more favourably located melanomas. In essence, in these centres the adverse outcomes of the more difficult tumours were diluted by the good results that can be obtained from protons used to treat tumours that can be more simply treated by plaque.

The side-effects on the external eye are difficult to quantify for the purposes of publication; however, they are easy to predict from the physics of the beam. We found that, when the lids and anterior segment were within the 90% isodose of the beam, these structures would always sustain severe complications whereas when they received only 50% of the tumour dose the effects were mild. It was an easy matter therefore to select only the patients who would have a good outcome from the point of view of surface problems. Similarly, we wanted to avoid subjecting patients to protons when they were likely to develop rubeosis. Even though some centres had indicated that rubeotic glaucoma is treatable provided that the tumour is also controlled, a quality of life study that we performed using an instrument of our own design¹⁶ showed that these were the only individuals amongst all our melanoma patients who regretted having a particular treatment. They were less satisfied than enucleation patients.

We were less sure of exactly which individuals are at greatest risk of rubeotic glaucoma and this was therefore the subject of our first proton study.¹⁷ This analysis comprised a prospective, consecutive series of 127 proton patients with a follow up of 3 years (range 2–6 years). As we already knew that large tumour size and persistent subretinal fluid were associated with rubeosis we paid special attention to these features. Because the goal of the study was to determine which melanomas would be better treated by enucleation, we defined large tumours as those that were considered too big for plaque radiotherapy. The remaining tumours that were small enough for plaque were mainly those that were regarded as being too close to the optic disc or macula for brachytherapy. The findings of the study are summarized in Table 6. They confirm our impression that rates of rubeotic glaucoma following proton therapy are high in patients with tumours too large to plaque and/or with retinal detachment. More than one in four patients with rubeosis (27.5%) required enucleation and most of the remainder required lengthy supportive therapy and had an uncomfortable, unsightly eye. Only the patients with tumours small enough

Table 7. Comparison of ruthenium plaque, iodine plaque and proton beam radiotherapy selected according to size and location of tumour (median follow up 4 years for whole series)

Treatment	No. eyes retained (%)
Ruthenium-106	125/140 (89)
Iodine-125	182/190 (95)
Protons	254/267 (96)

to plaque and no retinal detachment, mainly juxtapapillary melanomas, had low rates of rubeotic glaucoma and a good quality of life after protons.

Comparison of plaques and protons

Thus, by 1998 we were well on the way to answering the second question and ready to publish the results of 597 primary radiation treatments using ruthenium plaques, iodine plaques or protons, selected according to the size and location of the tumour.¹⁸ These are summarized in Table 7 and show how similar the results of iodine plaque and proton therapy were in terms of preservation of the eye. This series antedated our policy of dose modification. Consequently, ruthenium plaques lagged slightly behind in terms of achieving local tumour control.

CONCLUSIONS

By 2001 a formal prospective investigation had once and for all answered the first question. The COMS confirmed the findings of earlier retrospective studies that plaque radiotherapy is a safe alternative to enucleation although it had cost 660 enucleated eyes to prove the point. Meanwhile, the high complication rates of iodine plaque and proton treatments had encouraged us to persist with ruthenium plaques in London. We were able to show that, using bigger plaques and titrating the depth dose according to the thickness of the tumour, we could abolish the 6–7% inferiority of the local tumour control rate compared with protons and iodine plaques without significant loss of the advantage of the beta applicators in terms of reduced side-effects.

Following our evaluation of the risk factors predictive of rubeosis after proton therapy we abandoned this method for all except eyes with small tumours close to the disc, preferring to recommend enucleation for larger melanomas. This policy was based on claims that protons achieve better radiation dose distribution and therefore improved local

tumour control than plaques for melanomas close to the optic disc. These assertions have not been substantiated and we are currently trying to determine whether they can be justified. If they cannot, and if plaques achieve similar local tumour control rates to protons with equivalent or less side-effects, there can be little if any indication for proton therapy, which is at least 10 times as expensive as plaque radiotherapy in the UK.

Currently, we use ruthenium plaques for almost all tumours. We no longer employ iodine plaques at all and hardly ever recommend protons. If we find no significant difference between protons and plaques for juxtapapillary melanomas we may find ourselves using only ruthenium plaques!

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